

LAB 1: DETERMINATION OF REFRACTIVE INDEX VIA SNELL'S LAW

1 OBJECTIVES

- To understand the relationship between the angle of incidence (θ_1) and the angle of refraction (θ_2) at a plane boundary between two media.
- To experimentally determine the refractive index (n_2) of an unknown mystery material using Snell's Law.
- To identify the mystery material by matching the calculated refractive index against a database of known physical materials.
- To analyze experimental sources of error associated with optical protractor measurements.

2 THEORETICAL BACKGROUND

When a light ray travels from one transparent medium into another, its velocity changes due to differences in the optical density of the materials. This change in speed causes the ray to bend, a phenomenon known as refraction.

The relationship governing this bending is described by Snell's Law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

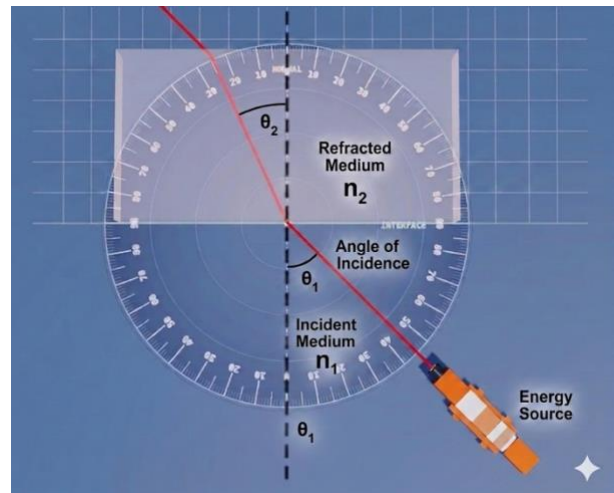
Where:

- n_1 is the refractive index of the incident medium (dimensionless).
- θ_1 is the angle of incidence, measured from the normal line perpendicular to the interface.
- n_2 is the refractive index of the refracting medium (dimensionless).
- θ_2 is the angle of refraction, measured from the normal line inside the second medium.

If the incident medium is air or a vacuum ($n_1 \approx 1.000$), Snell's Law simplifies to:

$$n_2 = \frac{\sin \theta_1}{\sin \theta_2}$$

By measuring multiple pairs of (θ_1, θ_2) and plotting $\sin \theta_1$ versus $\sin \theta_2$, we can determine n_2 with high precision from the slope of the linear regression line.



- Interactive Refraction Lab Simulator ([refraction_lab.html](#))
- Incident Medium (*Medium 1*): Air / Vacuum ($n_1 = 1.000$)
- Refracted Medium (*Medium 2*): Selected "Mystery Material" ($n_2 = ???$)
- Light Source: Laser diode set to any wavelength
- Measurement Aids: Virtual protractors (integrated circular and moveable mini-protractor), zoom loupe magnifier.

4 EXPERIMENTAL PROCEDURE

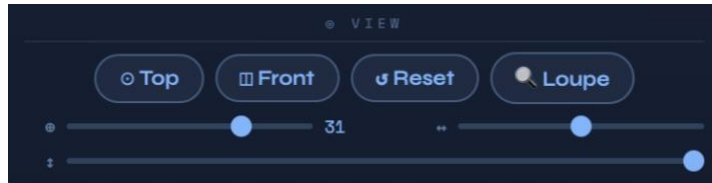
Part A: Entering Challenge Mode & Setting Up the System

- Open the virtual simulator [refraction_lab.html](#) in your web browser.
- Locate the Refracted Medium dropdown menu in the top control panel.
- Select 🎲 Mystery Material from the option list. This automatically engages Challenge Mode. Notice that the digital display of n_2 is hidden ($n_2 = ???$).

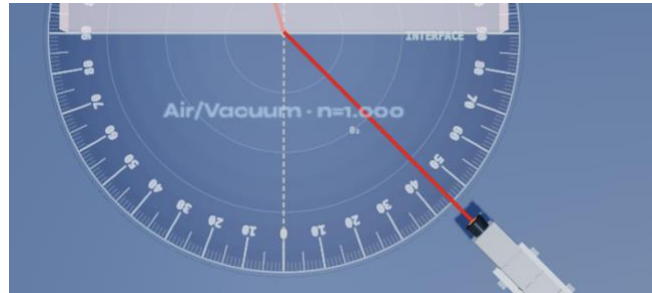


- Keep the Incident Medium set to Air / Vacuum ($n_1 = 1.000$).

1. In the navigation panel, click  Top to position the virtual camera looking straight down at the interface. This provides a clear, undistorted 2D view of the protractor grid.



2. Use the default Protractor to verify alignments on the primary circular scale.



3. Select your first angle of incidence (θ_1) by dragging the laser slider or manually aiming the laser starting with 20.0°.
4. Read the corresponding angle of refraction (θ_2) on the lower half of the scale.
 - *Tip:* Use the  Loupe tool to magnify the region where the refracted green beam intersects the protractor graduations to achieve precision to the nearest 0.1°.
5. Record both θ_1 and θ_2 in Table 1.1.
6. Repeat steps 3 to 5 for different angles of incidence.

5 SCREENSHOT SUBMISSION PLACEHOLDER

[PLACE KEY SCREENSHOT HERE]

Capture your simulation workspace in "Top" view showing the incident through the Mystery Material, with the Loupe tool highlighting the θ_2 intersection on the protractor scale.

6 DATA ACQUISITION & ANALYSIS



Use the recorded angular data to compute the sine functions and determine the experimental refractive index.

Table 1.1: Refraction Measurements and Calculation of n_2

- Constant: Refractive Index of Incident Medium (n_1) = 1.000 (Air)

Trial	Incident Angle θ_1 (deg)	Refracted Angle θ_2 (deg)	Calculated $n_2 = \frac{\sin \theta_1}{\sin \theta_2}$
1	20.0°		
2	45.0°		
3	60.0°		
4	80.0°		
Average:			Mean $n_{2, \text{exp}} = \underline{\hspace{2cm}}$

7 MATERIAL IDENTIFICATION

Below is a reference index table of known optical materials. Compare your calculated mean refractive index ($n_{2, \text{exp}}$) to find the closest match.

Table 1.2: Standard Reference Material Index List

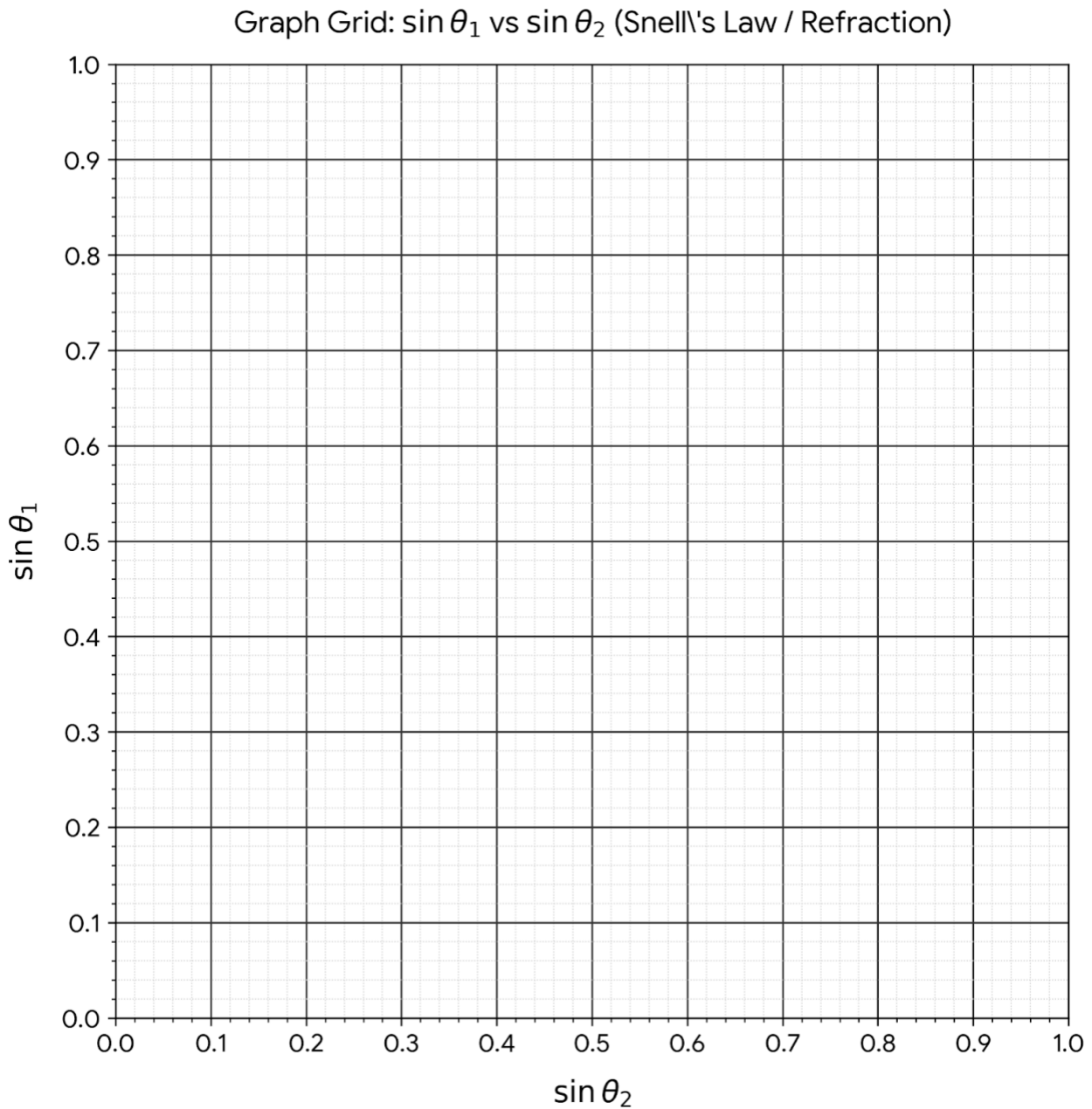
Material	Refractive Index (n)	Material	Refractive Index (n)
Fluorite	1.43	Flint Glass	1.65
Fused Quartz	1.46	Sapphire	1.77
Crown Glass	1.52	Cubic Zirconia	2.15
Polycarbonate	1.59	Gallium Phosphide	3.05

- Identified Mystery Material: _____
- Reference Refractive Index ($n_{2, \text{ref}}$): _____
- Percentage Error Calculation:

$$\% \text{ Error} = \left| \frac{n_{2, \text{exp}} - n_{2, \text{ref}}}{n_{2, \text{ref}}} \right| \times 100\% = \underline{\hspace{2cm}}\%$$

8 DISCUSSION & CONCLUSION

1. Linear Analysis: Plot a graph of $\sin \theta_1$ (y-axis) versus $\sin \theta_2$ (x-axis) on graph paper or a graphing tool. Sketch the line of best fit. Explain why the slope of this line represents the experimental refractive index of the mystery medium.



2. Precision & Errors: Identify at least two systematic or random errors that might arise when taking measurements with the virtual protractor. How did utilizing the Loupe tool help mitigate these errors?